

Generation Interconnection System Impact Study Report

For Queue Project Q19-1175

Roosevelt Dam 138 kV

January 31, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1175. The Interconnection Customer proposes a 144.6 MW solar plus storage facility interconnecting at the Roosevelt Dam 138 kV substation in the New Jersey area. The request was received on March 25, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 328 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,889,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Yahara Riv - Colorado 138 KV	Russell Energy Ctr - Fair Lawn 138 KV	144.0	271	6.2
Russell Energy Ctr - Cabinet Gorge 138 KV line	Romn Nose - Parsons 138 KV line	120.8	235	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yahara Riv - Colorado 138 KV	Rebuild line with higher-capacity conductor	\$502,000	10% (\$51,000)
Russell Energy Ctr - Cabinet Gorge 138 KV line	Rebuild line with higher-capacity conductor	\$3,838,000	100% (\$3,838,000)
	Total allocated to project		\$3,889,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.